NITHIN REDDY THUMMALA

Azure Snowflake Data Engineer

Email:nithinreddy080319@gmail.com**Phone**:(469) 213-8807**LinkedIn**:https://www.linkedin.com/in/nithin-reddy-thummala-557b71187/

PROFESSIONAL SUMMARY

- To obtain a challenging Data Engineer position that leverages my 11+ years of experience in the software industry, including 5+ years of experience in Azure cloud services and big data technologies, and 5+ years of experience in Data warehouse implementations.
- Expertise in designing and implementing scalable data ingestion pipelines using Azure Data Factory. Proficient in leveraging Azure Databricks and Spark for distributed data processing and transformation tasks.
- Adept at designing cloud-based data warehouse solutions using Snowflake on Azure, optimizing schemas, tables, and views for efficient data storage and retrieval.
- Experience in Migrating SQL database to Azure Data Lake, Azure SQL Database, Databricks, and Azure SQL Data warehouse.
- Excellent Experience of integrating Azure Data Factory V2/V1 with various data sources and processing the data using the pipelines, pipeline parameters, activities, activity parameters, and manual/window based/event-based job scheduling.
- Have experience designing and developing Azure stream analytics jobs to process real-time data using Azure Event Hubs. Strong expertise in optimizing Spark jobs and leveraging Azure Synapse Analytics for big data processing and analytics.
- Implemented data pipelines using snow SQL, Snowflake Integrated services, and snow pipe.
- Participates in the development, improvement, and maintenance of snowflake database applications. Build the Logical and Physical data model for snowflake as per the changes required.
- Define roles and privileges required to access different database objects. In-depth knowledge of Snowflake Database, Schema and Table structures.
- Proficient in scripting languages such as Python and Scala. Developed UNIX shell scripts for Hadoop & Big Data Development.
- Good working experience in using Spark SQL to manipulate Data Frames in Python. Skilled in working with Hive, Spark SQL, Kafka, and Spark Streaming for ETL tasks and real-time data processing.
- I am experienced in setting up workflow using Apache Oozie workflow engine for managing and scheduling Hadoop jobs and importing and exporting data using SQOOP from HDFS to Relational Database systems.
- I have optimized query performance in Hive using bucketing and partitioning techniques and have extensive hands-on experience tuning spark Jobs.
- Highly proficient in Agile methodologies, including JIRA for project management and reporting. Experienced in Advance PL/SQL using Cursor, triggers, stored procedures, views, materialized views, functions for data integrity and enforcing business rules.

EDUCATION

- Bachelor’s in computer science engineering from Jawaharlal Nehru Technological University, India.

TECHNICAL SKILLS

Azure Services	Azure data Factory, Azure Data Bricks, snowflake, Logic Apps, Functional App, Snowflake, Azure DevOps

Big Data Technologies	MapReduce, Hive, Teg, Python, PySpark, Scala, Kafka, Spark streaming, Oozie, Sqoop, Zookeeper, AWS
Hadoop Distribution	Cloudera, Horton Works
Languages	SQL, PL/SQL, Python, HiveQL, Scala.
Web Technologies	HTML, CSS, JavaScript, XML, JSP, Restful, SOAP
Operating Systems	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.
Build Automation tools	Ant, Maven
Version Control	GIT, GitHub.
IDE &Build Tools, Design	Eclipse, Visual Studio.
Databases	MS SQL Server 2016/2014/2012, Azure SQL DB, Azure Synapse. MS Excel, MS Access, Oracle 11g/12c, Cosmos DB

WORK EXPERIENCE

Client: American Express, Phoenix, AZ| Jun 2023 to till date
Role: Azure Snowflake data engineer
Responsibilities:

- Designed and implemented scalable data ingestion pipelines using Azure Data Factory, ingesting data from various sources such as SQL databases, CSV files, and REST APIs.
- Developed data processing workflows using Azure Databricks, leveraging Spark for distributed data processing and transformation tasks. Developed Spark core and Spark SQL scripts using Scala for faster data processing.
- Implemented Azure copy activity and dataflow jobs using azure data factory. Ensured data quality and integrity by performing data validation, cleansing, and transformation operations using Azure Data Factory and Databricks.
- Designed and implemented a cloud-based data warehouse solution using Snowflake on Azure, leveraging its scalability and performance capabilities.
- Created and optimized Snowflake schemas, tables, and views to support efficient data storage and retrieval for analytics and reporting purposes.
- Collaborated with data analysts and business stakeholders to understand their requirements and implemented xappropriate data models and structures in Snowflake.
- Developed and optimized Spark jobs to perform data transformations, aggregations, and machine learning tasks on big data sets.
- Implemented data lineage and metadata management solutions to track and monitor data flow and transformations.
- Identified and resolved performance bottlenecks in data processing and storage layers, optimizing query execution and reducing data latency.
- Implemented **DBT (Data Build Tool)** to transform and model data directly in Snowflake, enabling modular and maintainable data transformation pipelines.
- Designed and deployed DBT models to standardize business logic, ensuring consistency across analytics and reporting layers.
- Leveraged DBT's Jinja templating to dynamically generate SQL queries, improving redundancy and improving code reusability.
- Integrated DBT with Azure DevOps for CI/CD pipelines, automating testing and deployment of data models
- Utilized DBT documentation and testing features to ensure data quality, validating transformations with automated tests.
- Implemented partitioning, indexing, and caching strategies in Snowflake and Azure services to enhance query performance and reduce processing time.

- Collaborated with DevOps engineers to develop automated CI/CD and test-driven development pipeline using azure as per the client requirement.
- Successfully integrated Power BI with a variety of data sources, including on-premises databases, cloud services, and REST APIs, ensuring real-time or scheduled data updates.
- Customized Power BI reports using themes, color palettes, and templates to match the branding and styling preferences of the organization.
- Performed data transformation and cleansing within Power BI using the Power Query Editor, ensuring data accuracy and consistency.
- Identified and resolved data quality issues within incoming data feeds by implementing data cleansing and validation routines in SSIS, leading to a 20% reduction in data anomalies.
- Contributed to the development of a comprehensive data warehousing solution by designing SSIS packages to extract, transform, and load data from various sources into the data warehouse.
- Collaborated on ETL tasks, maintaining data integrity, and verifying pipeline stability. Document ETL processes, data mappings, data lineage, and workflows for knowledge sharing and future reference.
- Developed a data pipeline using Kafka, Spark, and Hive to ingest, transform and analyse data. Designed and implemented real-time data processing solutions using Kafka and Spark Streaming, enabling the ingestion, transformation, and analysis of high-volume streaming data.

Client: Microsoft, Seattle, Washington| Jan 2022 to April 2023
Role: Azure Snowflake data engineer
Responsibilities:

- Implemented end-to-end data pipelines using Azure Data Factory to extract, transform, and load (ETL) data from diverse sources into Snowflake.
- Built scalable and optimized Snowflake schemas, tables, and views to support complex analytics queries and reporting requirements.
- Developed data ingestion pipelines using Azure Event Hubs and Azure Functions to enable real-time data streaming into Snowflake.
- Leveraged Azure Data Lake Storage as a data lake for storing raw and processed data, implementing data partitioning and data retention strategies.
- Integrated SSRS reports with the data warehouse, providing users with unified access to up-to-date and accurate business insights.
- Utilized Azure Blob Storage for efficient storage and retrieval of data files, implementing compression and encryption techniques to optimize storage costs and data security.
- Implemented data governance practices and data quality checks using Azure Data Factory and Snowflake, ensuring data accuracy and consistency.
- Adopted DBT Core to build scalable data transformation workflows in Snowflake, replacing manual SQL scripts with version-controlled, modular pipelines.
- Created DBT macros to encapsulate complex business logic, simplifying maintenance and enhancing collaboration across teams.
- Implemented incremental DBT models to optimize performance for large datasets, reducing compute costs in Snowflake.
- Set up DBT snapshots to track historical changes in dimension tables, supporting Type 2 Slowly Changing Dimensions (SCD).
- Used DBT's dependency graph to visualize data lineage, improving governance and impact analysis for downstream reports.
- Integrated DBT with Power BI via semantic layers, ensuring consistent metrics for dashboards and self-service analytics.

- Implemented data replication and synchronization strategies between Snowflake and other data platforms using Azure Data Factory and Change Data Capture techniques.
- Designed and implemented data archiving and data retention strategies using Azure Blob Storage and Snowflake's Time Travel feature.
- Developed custom monitoring and alerting solutions using Azure Monitor and Snowflake Query Performance Monitoring (QPM) for proactive identification and resolution of performance issues.
- Integrated Snowflake with Power BI and Azure Analysis Services for creating interactive dashboards and reports, enabling self-service analytics for business users.
- Visualized data and analysis results in Databricks notebooks using libraries like Matplotlib or integrated third-party BI tools like Power BI.
- Created comprehensive documentation on Power BI dashboard design, data sources, and business logic, enabling easy knowledge transfer and maintenance.
- Utilized the Great Expectations python library to run data validations on raw data ingested in the s3 buckets before moving them into Redshift and used a Multi-node Redshift data warehouse to implement Columnar Data Storage, Advanced Compression, and Massive Parallel Processing.
- Collaborated with cross-functional teams including data scientists, data analysts, and business stakeholders to understand data requirements and deliver scalable and reliable data solutions.

Client: JB Hunt Transportation, Bentonville, AR |Mar 2019 to Dec 2021
Role: Big data Developer
Responsibilities:

- Designed comprehensive data ingestion pipelines using Sqoop, Flume, Kafka, and NiFi, enabling real-time processing and analysis of customer behavioral data.
- Implemented robust data governance through NiFi, emphasizing data lineage and metadata management in line with regulatory standards.
- Utilized Apache Spark and Scala to aggregate and analyze vast datasets, driving transformative business insights.
- Orchestrated end-to-end data solutions in Azure, focusing on ETL operations via Azure Data Factory and SSIS, boosting data storage and integration.
- Crafted insightful dashboards and analytics using Power BI, powered by Python and Scala scripts, enhancing data-driven decision-making for stakeholders.
- Conducted data transformation and feature engineering tasks within Databricks notebooks, preparing data for machine learning and analytics.
- Utilized Databricks for machine learning and AI model development, training models, and deploying them for real-time scoring or batch processing.
- Developed an Azure-based alert mechanism leveraging Databricks, Spark, and Azure Data Factory for efficient data processing across various Azure services.
- Integrated HBase with Hive, optimizing table structures for swift data retrieval and efficient querying.
- Streamlined database management in Kubernetes, devising strategies for backup, restoration, and scalability, ensuring consistent data availability.
- Collaborated with the IT team to ensure seamless integration of Power BI with existing enterprise systems using Power BI Gateways.
- Automated the refresh schedules for Power BI reports using Power BI service, ensuring stakeholders always had access to the latest data insights.

- Authored diverse SQL scripts for tasks ranging from data migration to analytics, employing ANSI SQL standards for broad compatibility.
- Leveraged Hive queries and Spark SQL, enhanced with Python and Scala, for versatile and efficient data processing.
- Automated deployment processes using YAML, ensuring swift and reliable builds and release cycles.
- Championed the migration of datasets from Oracle RDBMS to Hadoop using Sqoop, enhancing overall data processing capabilities.
- Employed JIRA for project management, enhancing issue tracking, and workflow organization.
- Collaborated intensively with the team to troubleshoot JVM-related challenges, reinforcing system stability and performance.
- Maintained and managed source code using Git, facilitating team collaboration and efficient version control.

Client: State of OH, Columbus, OH| Mar 2016 to Nov 2018
Role: Big data Developer
Responsibilities:

- Designed a comprehensive ETL framework with Sqoop, Pig, and Hive, enabling efficient data ingestion from various sources for consumption.
- Leveraged Spark (RDDs, Data Frames, Spark SQL) and Spark-Cassandra Connector for versatile tasks, such as data migration from Oracle to MySQL and business report generation.
- Developed real-time sales analytics using Spark Streaming, catering to dynamic business requirements.
- Played a pivotal role in migrating data from on-premises SQL server to Azure Synapse Analytics & Azure SQL DB, enhancing cloud capabilities and storage efficiency.
- Utilized Hive to process vast HDFS datasets, creating external tables, and establishing scripts for table ingestion that enhanced reusability across projects.
- Integrated Databricks with various Azure services, such as Azure Synapse Analytics (formerly SQL Data Warehouse), Azure Data Factory, and Azure DevOps for end-to-end data workflows.
- Expertise in big data ecosystems, having worked with tools such as Apache Hive, Pig, HBase, Spark, Zookeeper, Flume, Kafka, and Sqoop for diverse data engineering tasks.
- Optimized MapReduce job performances using combiners, partitioners, and distributed cache, ensuring efficient data processing at scale.
- Automated deployments with YAML scripts, streamlining build and release cycles for improved development efficiency.
- Managed source code and ensured collaborative best practices using Git and GitHub, facilitating efficient version control and team collaborations.
- Analyzed source data, managed datatype conversions, and utilized Power BI to produce ad-hoc reports, catering to dynamic analytical requirements.
- Utilized Power Query in Power BI for intricate data transformations, cleaning, and data preparation tasks, enabling cleaner and more accurate visualizations.
- Implemented row-level security (RLS) within Power BI dashboards, ensuring data access was role-specific and compliant with company data policies.
- Collaborated cross-functionally, translating business requirements into actionable data solutions using Azure tools like Data Lake, Data Factory, and SQL Database.

Client: HCA, Nashville, TN |Oct 2013 to Jan 2016
Role: Big data Developer

Responsibilities:

- Assisted in the design and deployment of big data solutions using Hadoop ecosystem tools such as Hive, Sqoop, Pig, Flume, Oozie, and Kafka in integration with Azure's cloud infrastructure.
- Played an active role in data ingestion from various sources into Hadoop clusters using Sqoop and Flume and orchestrated these workflows using Oozie.
- Utilized Hive for querying and managing large datasets residing in distributed storage, while seamlessly integrating with Azure Data Lake for enhanced data storage and retrieval.
- Conducted performance tuning in Databricks to optimize Spark jobs, enhance cluster performance, and reduce data processing times.
- Collaborated with data engineering and data science teams within Databricks workspaces, using Git or other version control systems for collaborative development.
- Explored real-time data streaming capabilities using Kafka, ensuring data is immediately available for processing and analytics in both Hadoop and Azure environments.
- Supported the development of ETL processes using Pig, Python, and Scala to transform data and load it into desired storage solutions within Azure and Hadoop ecosystems.
- Assisted in the configuration and maintenance of Spark clusters, both in Hadoop and Azure Databricks environments, and wrote basic Spark scripts using Python (PySpark) and Scala.
- Gained hands-on experience in Azure's native big data solutions, particularly Azure HDInsight, understanding its compatibility and integration with the Hadoop ecosystem.
- Worked closely with senior engineers to ensure data security and compliance across both Hadoop and Azure platforms, understanding the criticality of encryption, role-based access controls, and managed identities.
- Actively participated in team meetings and training sessions, grasping the evolving landscape of big data and cloud technologies, and understanding the synergy between Hadoop tools and Azure services.
- Collaborated with cross-functional teams, ensuring data is processed and available for data scientists, analysts, and other stakeholders in a timely and efficient manner.
- Embraced a continuous learning approach, regularly attending workshops and webinars to stay updated with advancements in Hadoop, Azure, Python, Scala, and associated technologies.

Client: Sky Bridge Global Pvt. Ltd, India | Aug 2012 to Sep 2013
Role: Data Warehouse Developer
Responsibilities

- Create and maintain databases for Server Inventory, Performance Inventory.
- Worked in Agile Scrum Methodology with daily stand-up meetings, great knowledge working with Visual SourceSafe for Visual studio 2010 and tracking the projects using Trello.
- Generated Drill through and Drill down reports with Drop down menu option, sorting the data, and defining subtotals in Power BI.
- Led the design and development of ETL processes using Informatica PowerCenter to extract, transform, and load data from various source systems to the data warehouse.
- Developed complex mappings, mapplets, and reusable transformations to ensure efficient data transformation and validation.
- Integrated Power BI with various data sources, including SQL databases, Excel spreadsheets, cloud-based storage, and third-party APIs to create real-time visualizations.
- Used Data warehouse for developing Data Mart which for feeding downstream reports, development of User Access Tool using which users can create ad-hoc reports and run queries to analyse data in the proposed Cube.

- Analysed the system for new enhancements/functionalities and performed Impact analysis of the application for implementing ETL changes. Built performant, scalable ETL processes to load, cleanse and validate data.
- Wrote a MapReduce job using Pig Latin. Involved in ETL, Data Integration and Migration.
- Created logical and physical designs of the database and ER Diagrams for Relational and Dimensional databases using Erwin.
- Deployed the SSIS Packages and created jobs for efficient running of the packages.
- Expertise in creating ETL packages using SSIS to extract data from heterogeneous database and then transform and load into the data mart.
- Involved in creating SSIS jobs to automate the reports generation, cube refresh packages.
- Thorough knowledge of Features, Structure, Attributes, Hierarchies, Star and Snowflake Schemas of Data Marts.
- Experienced with SQL Server Reporting Services (SSRS) to author, manage, and deliver both paper-based and interactive Web-based reports.
- Developed stored procedures and triggers to facilitate consistent data entry into the database.